

Coaching Institute Management System (CIMS)

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Version: 1.0

Document Status: Draft

Prepared By: Software Engineering Team

Project Name: Coaching Institute Management System (CIMS)

Document Type: Software Requirements Specification (SRS)

Classification: Internal Development Document

Date: July 2026

Coaching Institute Management System (CIMS)- RAMP Project

Table of Contents

Cover Page

Revision History

Table of Contents

1. Introduction

1.1 Purpose

1.2 Project Overview

1.3 Existing System

1.4 Proposed System

1.5 Problems in Existing System

1.6 Expected Benefits

1.7 Definitions

1.8 Acronyms

1.9 References

1.10 Document Conventions

2. System Overview

2.1 System Vision

2.2 System Features

2.3 System Architecture Overview

2.4 Major Modules

2.5 Deployment Overview

2.6 Technology Overview

3. Overall Description

3.1 Product Perspective

3.2 Product Functions

3.3 User Characteristics

3.4 Operating Environment

3.5 Design Constraints

3.6 User Documentation

3.7 Assumptions and Dependencies

4. System Features

4.1 Student Management

4.2 Faculty Management

4.3 Parent Portal

4.4 Attendance Management

4.5 Examination Management

4.6 Fee Management

4.7 Timetable Management

4.8 Reports & Analytics

4.9 Notifications

Coaching Institute Management System (CIMS)- RAMP Project

4.10 Administration

5. Executive Summary

6. Business Background

7. Problem Statement

8. Current System Analysis

9. Proposed Solution

10. Business Objectives

11. Scope of Work

12. Stakeholder Analysis

13. User Roles & Responsibilities

14. Functional Requirements

14.1 Authentication & Identity Management

14.2 User & Role Management

14.3 Institute Management

14.4 Branch Management

14.5 Student Management

14.6 Parent Portal

14.7 Course Management

14.8 Batch Management

14.9 Timetable Management

14.10 Attendance Management

14.11 Face Recognition Attendance

14.12 Fingerprint Attendance

14.13 Examination Management

14.14 Assignment & Homework Management

14.15 Study Material Management

14.16 Fee Management

14.17 Online Payment Integration

14.18 Notification & Communication Management

14.19 Dashboard & Business Analytics

14.20 Reports Management

14.21 Certificate Management

14.22 Artificial Intelligence Features (Future Ready)

15. Business Rules

16. Validation Rules

17. Non-Functional Requirements

17.1 Performance Requirements

17.2 Scalability Requirements

17.3 Availability Requirements

17.4 Reliability Requirements

Coaching Institute Management System (CIMS)- RAMP Project

17.5 Usability Requirements

18. System Architecture

18.1 Architectural Overview

18.2 Technology Stack

18.3 Deployment Architecture

19. Database Design

19.1 Database Overview

19.2 Database Entities

19.3 Entity Relationships

20. API Requirements

20.1 Authentication APIs

20.2 Student APIs

20.3 Attendance APIs

20.4 Fee APIs

20.5 Examination APIs

20.6 Report APIs

21. Security Requirements

22. Audit Requirements

23. Integration Requirements

24. Backup & Disaster Recovery

25. Data Migration Strategy

25.1 Migration Objectives

25.2 Migration Phases

25.3 Supported Import Formats

26. Acceptance Criteria

26.1 Functional Acceptance

26.2 Performance Acceptance

26.3 Security Acceptance

27. User Acceptance Testing (UAT)

28. Risk Analysis

29. Assumptions & Constraints

29.1 Assumptions

29.2 Constraints

30. Training Requirements

31. Warranty & Maintenance

31.1 Preventive Maintenance

31.2 Corrective Maintenance

31.3 Adaptive Maintenance

31.4 Perfective Maintenance

32. Future Enhancements

Coaching Institute Management System (CIMS)- RAMP Project

33. Requirements Traceability Matrix (RTM)

34. Appendices

Appendix A – Recommended Hardware Requirements

Appendix B – Recommended Software Stack

Appendix C – Glossary

35. Document Control

35.1 Document Approval

35.2 Document Control Information

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the functional, non-functional, technical, and operational requirements for the **Coaching Institute Management System (CIMS)**.

The document serves as the primary reference for developers, software architects, UI/UX designers, testers, project managers, institute administrators, and stakeholders throughout the software development lifecycle.

This SRS defines how the Coaching Institute Management System should operate, the services it should provide, the constraints under which it must function, and the expected quality standards. It establishes a common understanding among all stakeholders and ensures that the final product satisfies the operational needs of coaching institutions.

1.2 Project Overview

The Coaching Institute Management System (CIMS) is a comprehensive web-based and mobile-enabled institute management platform designed to digitize and automate the daily operations of coaching institutions.

Unlike traditional institute management software that focuses only on student records and fee collection, CIMS provides an end-to-end operational ecosystem for coaching centers by connecting administrators, teachers, students, parents, accountants, counselors, and institute owners through role-based secure access.

The system aims to eliminate manual paperwork, reduce administrative workload, improve operational transparency, provide real-time information, enhance communication between institutes and parents, and support data-driven decision-making through dashboards and analytical reports.

1.3 Existing System

Most coaching institutes currently manage their daily operations using a combination of manual registers, spreadsheets, WhatsApp groups, standalone accounting software, and paper-based documentation.

Student admission records are often maintained in physical files or Microsoft Excel sheets. Attendance is generally recorded manually by teachers in attendance registers, making it difficult to monitor attendance trends and identify irregular students.

Fee collection is frequently tracked using notebooks or spreadsheets, resulting in delayed fee reminders, inaccurate outstanding calculations, and increased administrative effort.

Batch scheduling is managed manually, which often causes classroom conflicts, faculty scheduling issues, and inefficient utilization of institute resources. Parents generally receive limited visibility regarding their child's attendance, fee status, examination performance, and academic progress.

Coaching Institute Management System (CIMS)- RAMP Project

Communication is mostly dependent on phone calls or messaging applications. Examination marks, assignments, student progress reports, and timetable changes are maintained separately without a centralized academic record.

1.4 Proposed System

The proposed Coaching Institute Management System introduces a fully integrated digital platform that automates every major operational activity within a coaching institute.

The system enables centralized management of student admissions, course enrollment, batch allocation, faculty management, attendance recording, fee collection, examination management, timetable scheduling, communication, reporting, and administrative operations.

Parents receive dedicated portal access through which they can monitor their child's attendance, fee payment history, examination results, progress reports, class schedules, announcements, homework, and institute notifications in real time.

Faculty members can manage attendance, conduct examinations, upload marks, distribute study materials, assign homework, communicate with students, and generate academic reports directly from the system.

Administrative staff can efficiently manage admissions, student records, fee collection, installment plans, scholarships, discounts, batch assignments, classroom scheduling, and institute operations through a unified dashboard.

1.5 Problems in Existing System

The current manual or semi-digital management approach creates several operational and administrative challenges for coaching institutions.

Student admission records are scattered across multiple physical registers and spreadsheet files, making information retrieval slow and inefficient.

Attendance records maintained manually are susceptible to manipulation, missing entries, duplicate records, and inaccurate reporting.

Fee collection processes rely heavily on manual calculations, increasing the likelihood of accounting errors, missed installments, incorrect balances, and delayed reminders.

Parents have limited visibility into their child's academic performance, attendance, examination results, and institute activities.

Manual batch scheduling frequently results in timetable conflicts, classroom overbooking, and faculty scheduling issues. Preparation of examination reports, attendance summaries, student performance analyses, and financial reports requires significant manual effort and consumes valuable administrative time.

Coaching Institute Management System (CIMS)- RAMP Project

Communication between institute management, teachers, students, and parents lacks a centralized platform, causing delays in sharing important announcements and updates.

As coaching institutes expand into multiple branches, managing operations across locations becomes increasingly difficult without a centralized information system.

Manual document management increases the risk of misplaced records, unauthorized access, and data loss. The absence of centralized analytics limits the ability of institute management to evaluate operational performance and make strategic decisions.

1.6 Expected Benefits

Implementation of the Coaching Institute Management System will significantly improve operational efficiency, transparency, academic management, and communication across the coaching institute ecosystem.

Student information will be securely stored within a centralized database, enabling instant retrieval and reducing administrative effort.

Digital attendance mechanisms such as facial recognition, fingerprint authentication, and authorized manual entry will improve attendance accuracy and reduce proxy attendance.

Automated fee management will streamline billing, installment tracking, receipt generation, overdue reminders, scholarship handling, and financial reporting.

Parents will gain continuous visibility into their child's attendance, academic performance, fee status, examination schedules, and institute announcements through a dedicated parent portal.

Faculty members will spend less time on administrative work and more time focusing on academic delivery through integrated attendance, examination, and course management tools.

Institute management will obtain comprehensive dashboards and reports covering admissions, finances, student performance, faculty workload, attendance trends, and operational metrics.

Centralized communication will improve collaboration between administrators, teachers, students, and parents through notifications, announcements, SMS, email, and mobile application alerts.

The system will support future expansion into multiple branches, additional courses, online classes, AI-based analytics, and advanced educational technologies without requiring major architectural changes.

1.7 Definitions

Term	Definition
Institute	Coaching organization operating one or more educational branches.
Branch	A physical coaching center managed under the institute.

Coaching Institute Management System (CIMS)- RAMP Project

Student	An individual enrolled in one or more coaching courses.
Parent	Guardian authorized to monitor student information through the parent portal.
Faculty	Teacher responsible for conducting classes, assessments, and attendance.
Batch	A scheduled group of students attending classes together.
Course	Educational program offered by the coaching institute.
Attendance	Daily record indicating student presence or absence in scheduled classes.
Fee	Amount payable by students toward enrollment in coaching programs.
Examination	Assessment conducted to evaluate student learning progress.
Timetable	Schedule defining class timings, classrooms, faculty assignments, and batch sessions.
Administrator	Authorized user responsible for institute configuration and operational management.

1.8 Acronyms

Acronym	Meaning
CIMS	Coaching Institute Management System
RBAC	Role-Based Access Control
API	Application Programming Interface
UI	User Interface
UX	User Experience
JWT	JSON Web Token
DBMS	Database Management System
OCR	Optical Character Recognition
KPI	Key Performance Indicator
SMS	Short Message Service
PDF	Portable Document Format
CSV	Comma-Separated Values
OTP	One-Time Password
UUID	Universally Unique Identifier

1.9 References

The system requirements and architectural decisions described in this document are based on:

- IEEE 29148 Software and Systems Engineering – Requirements Engineering
- ISO/IEC 25010 Software Quality Model
- OWASP Application Security Verification Standard (ASVS)
- REST API Design Best Practices
- Modern Web Application Architecture Principles
- Industry-standard Coaching Institute Operational Workflows
- Academic Institution Administration Best Practices

1.10 Document Conventions

This Software Requirements Specification follows a structured numbering convention to ensure consistency and traceability throughout the document.

Functional requirements will be uniquely identified using requirement identifiers (for example, FR-AUTH-001 and FR-STUDENT-001). Non-functional requirements, business rules, validation rules, API specifications, database entities, and test cases will use similar structured identifiers.

All requirements described in this document are considered mandatory unless explicitly identified as optional, configurable, or future enhancements.

The terms "shall", "must", and "required" indicate mandatory system behavior, while "may" and "optional" denote configurable or future capabilities.

2. System Overview

Coaching Institute Management System (CIMS)- RAMP Project

2.1 System Vision

The vision of the Coaching Institute Management System (CIMS) is to establish a unified, intelligent, and scalable digital platform that automates every major academic, administrative, operational, and financial activity within coaching institutions.

The system is designed to eliminate manual paperwork, reduce administrative overhead, improve communication among stakeholders, and provide real-time visibility into institutional operations. By integrating admissions, attendance, fee management, examinations, timetable scheduling, parent communication, and analytical reporting into a centralized platform, the system enables coaching institutes to operate more efficiently while delivering improved educational services.

CIMS aims to become a future-ready educational ecosystem capable of supporting Artificial Intelligence, Business Intelligence, biometric authentication, online learning, cloud computing, and mobile accessibility without requiring significant architectural modifications.

2.2 System Features

The Coaching Institute Management System provides a comprehensive set of integrated features that simplify institute operations while improving academic quality and operational transparency.

The system supports digital student admissions, centralized student information management, faculty management, parent portal access, attendance recording through Face Recognition, Fingerprint Authentication, and Manual Entry, fee management with online payment integration, examination scheduling, marks management, assignment distribution, timetable scheduling, study material management, automated notifications, analytical dashboards, certificate generation, report management, and role-based access control.

Additional capabilities include multi-branch management, audit logging, data backup, configurable workflows, customizable reports, AI-ready architecture, and integration with third-party services such as payment gateways, SMS gateways, Email services, biometric devices, and cloud storage.

2.3 System Architecture Overview

The Coaching Institute Management System follows a modern multi-layered architecture that separates presentation, business logic, data access, and infrastructure services.

The Presentation Layer consists of responsive web and future mobile applications that provide user interfaces for administrators, faculty, students, parents, accountants, and institute owners.

The Application Layer contains RESTful APIs and business services responsible for authentication, admissions, attendance, examinations, fee management, reporting, communication, and system administration.

The Integration Layer enables communication with external services such as payment gateways, SMS providers, Email services, Face Recognition engines, Fingerprint devices, cloud storage, and future Learning Management Systems.

2.4 Major Modules

The Coaching Institute Management System consists of multiple functional modules that work together as an integrated educational management platform.

Major modules include:

- Authentication & Identity Management
- User & Role Management
- Institute Management
- Branch Management
- Student Admission Management
- Student Information Management
- Parent Portal
- Faculty Management
- Staff Management
- Course Management
- Subject Management
- Batch Management
- Timetable Management
- Attendance Management
- Face Recognition Attendance
- Fingerprint Attendance
- Manual Attendance
- Examination Management
- Assignment Management
- Homework Management
- Study Material Management
- Fee Management
- Online Payment Management
- Notification Management
- Communication Management
- Dashboard & Analytics
- Reports Management
- Certificate Management
- Audit Management
- Backup & Recovery
- System Administration

Each module operates independently while sharing centralized data through a common database and standardized APIs.

2.5 Deployment Overview

The Coaching Institute Management System is designed for flexible deployment based on institutional requirements. The application may be deployed as a cloud-hosted Software-as-a-Service (SaaS) solution, an on-premises enterprise application, or a hybrid deployment model.

The web application is accessible through modern browsers including Google Chrome, Microsoft Edge, Mozilla Firefox, and Safari.

Future releases will include Android and iOS mobile applications for students, parents, faculty members, and administrators.

Cloud deployment supports automatic scalability, centralized monitoring, high availability, disaster recovery, automated backups, and continuous software updates.

2.6 Technology Overview

The Coaching Institute Management System is developed using modern technologies that ensure performance, scalability, security, and maintainability.

Layer	Technology
Frontend	React.js, TypeScript, Tailwind CSS
Backend	Node.js, Express.js
Database	PostgreSQL / MongoDB
Authentication	JWT, Refresh Tokens, OAuth
Cache	Redis
Cloud Storage	AWS S3 / Cloudinary
AI Engine	TensorFlow, FaceNet, OpenCV
Biometric Integration	Vendor SDK APIs
Mobile Application	Flutter (Future Release)
Reporting	PDF Generator, Excel Export
Deployment	Docker, Nginx
Cloud Platform	AWS, Azure, Google Cloud

3. Overall Description

3.1 Product Perspective

The Coaching Institute Management System is a standalone enterprise application developed to replace manual administrative processes and fragmented software solutions currently used by coaching institutes.

Coaching Institute Management System (CIMS)- RAMP Project

The system integrates all operational, academic, financial, and communication activities into a centralized platform accessible by authorized users through secure role-based authentication.

The application supports institutes of varying sizes, from single-branch coaching centers to large multi-branch organizations serving thousands of students.

3.2 Product Functions

The primary functions of the Coaching Institute Management System include:

- Student inquiry and admission management
- Student enrollment and batch allocation
- Course and subject management
- Faculty and staff management
- Attendance recording through Face Recognition, Fingerprint Authentication, and Manual Entry
- Timetable scheduling
- Examination management
- Assignment and homework management
- Fee management and online payment processing
- Parent portal access
- Student portal access
- Communication through SMS, Email, Push Notifications, and future WhatsApp integration
- Certificate generation
- Analytical dashboards
- Report generation
- Audit logging
- Backup management
- System administration

3.3 User Characteristics

The system supports multiple categories of users with different responsibilities and access permissions.

Primary user groups include:

- Super Administrator
- Institute Owner
- Managing Director
- Branch Manager
- Administrator
- Accountant
- Admission Counselor
- Reception Staff

Coaching Institute Management System (CIMS)- RAMP Project

- Faculty Member
- Student
- Parent
- IT Support Engineer

Each role is assigned specific permissions according to organizational responsibilities through Role-Based Access Control (RBAC).

3.4 Operating Environment

The Coaching Institute Management System operates within modern computing environments.

Supported environments include:

- Windows 10 and Windows 11
- Ubuntu Linux
- macOS
- Android (Future)
- iOS (Future)

Supported browsers include:

- Google Chrome
- Microsoft Edge
- Mozilla Firefox
- Safari

Database support includes PostgreSQL and MongoDB.

Cloud deployment is supported on AWS, Azure, and Google Cloud Platform.

3.5 Design Constraints

The design and implementation of the Coaching Institute Management System shall comply with the following constraints:

- Secure authentication and authorization mechanisms.
- Data privacy regulations applicable to educational institutions.
- Compatibility with biometric hardware.
- Compatibility with multiple payment gateways.
- Internet connectivity requirements for cloud deployment.
- Browser compatibility.
- Mobile responsiveness.
- Scalability for future organizational growth.
- High availability requirements.

Coaching Institute Management System (CIMS)- RAMP Project

- Backup and disaster recovery requirements.

3.6 User Documentation

The system shall provide comprehensive documentation for all stakeholder groups.

Documentation includes:

- System Administrator Manual
- Faculty User Guide
- Student User Guide
- Parent User Guide
- Accountant User Guide
- Installation Guide
- API Documentation
- Database Documentation
- Backup & Recovery Manual
- Troubleshooting Guide
- Frequently Asked Questions (FAQ)

3.7 Assumptions and Dependencies

Successful operation of the Coaching Institute Management System depends upon several assumptions.

- Reliable internet connectivity is available for cloud-hosted deployments.
- Biometric devices are properly configured before attendance operations begin.
- Face Recognition cameras meet recommended specifications.
- Payment gateway accounts are active.
- SMS and Email providers are correctly configured.
- Cloud infrastructure remains available.
- Users possess appropriate access credentials.
- Institute administrators configure academic structures before admitting students.

4. System Features

4.1 Student Management

The Student Management module manages the complete student lifecycle, including inquiry registration, admissions, document verification, enrollment, profile management, academic records, attendance history, fee information, examination results, and certification.

4.2 Faculty Management

The Faculty Management module manages faculty profiles, subject allocation, timetable assignments, attendance, examination responsibilities, assignment creation, academic performance tracking, and faculty workload management.

4.3 Parent Portal

The Parent Portal provides secure access to student attendance, fee information, examination results, assignments, homework, academic progress, announcements, notifications, and communication with faculty.

4.4 Attendance Management

The Attendance Management module supports Artificial Intelligence Face Recognition, Fingerprint Authentication, QR Code Attendance (optional), RFID Attendance (future), and Manual Attendance Entry by authorized faculty or administrative staff.

4.5 Examination Management

The Examination Management module supports examination scheduling, question paper management, marks entry, grading, ranking, report card generation, and analytical performance reporting.

4.6 Fee Management

The Fee Management module automates fee structures, invoice generation, installment planning, scholarships, concessions, online payments, receipt generation, outstanding tracking, refunds, and financial reporting.

4.7 Timetable Management

The Timetable Management module schedules classes, classrooms, faculty, and batches while automatically detecting scheduling conflicts and notifying affected stakeholders of timetable changes.

4.8 Reports & Analytics

The Reports and Analytics module provides operational dashboards, attendance analytics, examination analytics, admission trends, fee collection reports, faculty performance reports, student progress reports, and business intelligence dashboards for institute management.

4.9 Notifications

The Notification module delivers automated communications through SMS, Email, Push Notifications, and future WhatsApp integration for attendance, fees, examinations, timetable changes, announcements, and emergency alerts.

4.10 Administration

The Administration module provides centralized control over institute configuration, user management, role-based permissions, branch administration, academic calendars, security policies, audit logs, backups, integrations, and system monitoring.

5. Executive Summary

The Coaching Institute Management System (CIMS) is an integrated, enterprise-grade digital platform designed to automate, streamline, and optimize every operational activity performed within coaching institutions. The system replaces traditional paper-based administration and disconnected software tools with a centralized, secure, scalable, and intelligent management platform.

The solution is designed for coaching institutes ranging from small educational centers with a few hundred students to large multi-branch organizations serving thousands of students across multiple locations.

The platform manages the complete student lifecycle beginning from inquiry generation and admission through course enrollment, batch allocation, attendance, examinations, fee collection, academic performance monitoring, communication, and certificate generation.

Coaching Institute Management System (CIMS)- RAMP Project

Faculty members receive dedicated dashboards for class management, attendance recording, timetable viewing, examination management, assignment distribution, student performance tracking, and communication with students and parents.

Administrative staff can efficiently manage admissions, student records, fee collection, discounts, scholarships, installment plans, transportation, library records (optional), hostel management (optional), and institute operations through centralized dashboards.

Institute owners and directors receive business intelligence dashboards displaying operational KPIs, admission trends, revenue analytics, pending fee collections, faculty workload, classroom utilization, attendance statistics, examination performance, and branch-wise profitability reports.

The platform follows a modular architecture allowing future integration with Learning Management Systems (LMS), AI-powered student performance prediction, online examinations, virtual classrooms, payment gateways, WhatsApp Business API, biometric devices, and government educational compliance systems.

6. Business Background

Educational coaching institutes have experienced rapid growth over the last decade due to increasing competition in entrance examinations, government recruitment, higher education admissions, competitive certifications, and skill development programs.

As student enrollments increase, administrative operations become significantly more complex. Traditional management methods involving registers, spreadsheets, notebooks, and standalone accounting software create operational inefficiencies, inaccurate records, delayed reporting, and communication gaps.

Most coaching institutes simultaneously manage multiple courses, multiple batches, multiple faculty members, different fee structures, installment plans, examination schedules, attendance registers, parent communications, and academic records. Manual coordination across these activities consumes significant administrative resources while increasing the probability of human error.

Parents increasingly expect transparency regarding their children's attendance, academic progress, examination results, fee payments, homework, and classroom participation. Existing manual systems cannot provide real-time visibility or instant notifications, leading to communication delays and reduced parental engagement.

Coaching Institute Management System (CIMS)- RAMP Project

Faculty members spend substantial time maintaining attendance registers, preparing examination reports, calculating student rankings, updating academic records, and communicating individually with students and parents. These administrative responsibilities reduce valuable teaching time.

Modern coaching institutes therefore require a comprehensive digital platform capable of integrating academic management, administration, finance, communication, and analytics into a single ecosystem that supports sustainable organizational growth.

The Coaching Institute Management System has been conceptualized to fulfill these evolving operational, academic, administrative, and business management requirements.

7. Problem Statement

The majority of coaching institutes continue to rely upon fragmented manual processes that are no longer capable of efficiently managing growing student populations and increasingly complex academic operations.

Student admission records are frequently maintained using paper applications and spreadsheet files, making information retrieval slow, inconsistent, and vulnerable to duplication.

Attendance registers maintained manually are susceptible to proxy attendance, missing entries, calculation errors, unauthorized modifications, and delayed reporting.

Fee management processes often depend upon manual calculations, handwritten receipts, notebooks, and independent accounting software. These disconnected systems increase the likelihood of incorrect balances, delayed installment tracking, inaccurate outstanding calculations, and missed payment reminders.

Batch scheduling and faculty allocation are generally coordinated manually, resulting in classroom conflicts, scheduling overlaps, underutilized infrastructure, and administrative confusion.

Parents rarely receive immediate access to attendance information, examination performance, homework, timetable modifications, fee status, or institute announcements, requiring frequent phone calls and physical visits.

As coaching institutes expand into multiple branches, maintaining operational consistency, centralized reporting, standardized academic processes, and unified student records becomes increasingly difficult.

Coaching Institute Management System (CIMS)- RAMP Project

The absence of centralized operational analytics prevents management from identifying trends related to admissions, student retention, revenue generation, faculty performance, classroom utilization, and institutional growth.

8. Current System Analysis

The existing operational environment within most coaching institutes consists of multiple independent systems and manual processes operating without centralized coordination.

Student admissions are processed using physical application forms followed by manual data entry into spreadsheet software. Student documents are stored in physical files, making retrieval difficult and increasing the possibility of document loss.

Attendance is generally recorded in classroom registers maintained by faculty members. Monthly attendance percentages are calculated manually and often become available only after significant delays.

Fee collection relies upon handwritten receipts, spreadsheet calculations, or standalone accounting applications that are not integrated with student academic records. As a result, administrators frequently perform duplicate data entry across multiple systems.

Timetable preparation is performed manually using spreadsheets or printed schedules, requiring frequent revisions whenever faculty availability, classroom allocation, or batch timings change.

Examination management involves manual preparation of mark sheets, rank calculations, performance reports, and student progress summaries. Faculty members spend considerable effort consolidating examination data before reports can be generated.

Communication with parents depends primarily upon telephone calls, SMS messages, WhatsApp groups, or printed circulars. Important announcements may not reach every parent consistently.

Administrative reporting requires manual compilation of information from various registers and spreadsheet files, delaying management decision-making.

Coaching Institute Management System (CIMS)- RAMP Project

Most institutes lack centralized monitoring of academic performance, financial operations, attendance analytics, faculty productivity, classroom occupancy, and branch-level performance.

Consequently, administrative productivity decreases as student enrollment grows, creating scalability limitations that directly affect institutional efficiency and service quality.

9. Proposed Solution

The proposed Coaching Institute Management System introduces a fully integrated cloud-enabled management platform that centralizes every operational, academic, administrative, and financial activity within the coaching institute ecosystem.

The platform maintains a centralized database containing student records, faculty information, academic schedules, attendance data, examination records, fee transactions, communication history, and administrative configurations.

The admission management module automates inquiry registration, student enrollment, document verification, course allocation, batch assignment, ID generation, and digital record maintenance.

Attendance management supports multiple authentication mechanisms including AI-powered facial recognition, biometric fingerprint verification, QR code scanning, RFID cards (optional), and authorized manual attendance entry. Attendance data becomes immediately available to administrators, faculty members, students, and parents according to role-based permissions.

Fee management automates invoice generation, installment planning, scholarship processing, concession approval, payment tracking, online payment integration, receipt generation, outstanding balance monitoring, and automated reminder notifications.

The examination management module supports examination scheduling, question paper management, mark entry, grading, ranking, performance analysis, subject-wise analytics, and report card generation.

Parents receive secure access to dedicated portals where they can continuously monitor attendance records, academic performance, examination results, homework, fee status, announcements, and institute communications.

The proposed platform establishes a unified digital ecosystem that improves transparency, operational efficiency, communication quality, academic management, financial accuracy, and long-term organizational scalability.

10. Business Objectives

The primary objective of the Coaching Institute Management System is to digitally transform coaching institute operations through automation, centralized information management, intelligent reporting, and enhanced stakeholder collaboration.

The system aims to eliminate manual administrative processes and reduce dependency on paper-based documentation by introducing secure digital workflows across all departments.

The platform seeks to improve operational efficiency by automating repetitive administrative activities including admissions, attendance, fee collection, examination management, timetable scheduling, report generation, and communication.

Another major objective is to increase transparency between coaching institutes and parents by providing real-time access to attendance records, academic performance, fee details, announcements, and student progress reports.

The system intends to strengthen academic quality through continuous performance monitoring, analytical dashboards, examination insights, and early identification of academically underperforming students.

The platform is designed to support institutional growth by enabling centralized management of multiple branches, courses, classrooms, faculty members, and student populations through a single administrative system.

The solution also aims to improve financial management through automated billing, fee tracking, online payment support, revenue analytics, and outstanding balance monitoring.

Future readiness is another strategic objective, ensuring that artificial intelligence, predictive analytics, online learning, virtual classrooms, and advanced educational technologies can be integrated without significant architectural modifications.

11. Scope of Work

The Coaching Institute Management System encompasses the complete operational lifecycle of coaching institute management beginning from student inquiry and admission through academic completion and certification.

The software includes comprehensive management of institute configuration, branch administration, user management, student admissions, course creation, batch scheduling, classroom allocation, faculty management, attendance recording, examination management, fee collection, communication, reporting, and analytics.

Attendance management includes facial recognition, biometric fingerprint authentication, manual attendance entry, QR code attendance (optional), attendance correction workflows, absentee monitoring, attendance analytics, and attendance notifications.

Academic management includes course planning, syllabus tracking, timetable scheduling, examination scheduling, assignment management, homework distribution, marks management, grading, ranking, and student progress analysis.

Financial management includes fee structure configuration, installment planning, concessions, scholarships, online payment integration, receipt generation, outstanding tracking, refunds, and financial reporting.

Communication services include SMS notifications, email alerts, push notifications, WhatsApp integration (future enhancement), parent messaging, faculty announcements, and institute circulars.

The system supports role-based access for institute owners, administrators, accountants, faculty members, counselors, reception staff, students, parents, and branch managers.

The scope also includes comprehensive dashboards, reporting services, audit logging, security controls, API integration capabilities, backup mechanisms, and future scalability for cloud deployment and mobile application support.

12. Stakeholder Analysis

Stakeholder	Role	Responsibilities	System Benefits
-------------	------	------------------	-----------------

Coaching Institute Management System (CIMS)- RAMP Project

Institute Owner	Business Owner	Monitor institute operations, revenue, branch performance, strategic decisions	Complete operational visibility and business analytics
Managing Director	Executive Management	Supervise institutional growth and policy implementation	Executive dashboards and organizational insights
Administrator	System Administration	Configure system settings, users, permissions, institute operations	Centralized operational management
Branch Manager	Branch Operations	Manage branch-level admissions, faculty, students, and daily activities	Branch-specific administration and reporting
Admission Counselor	Student Admissions	Handle inquiries, admissions, counseling, and enrollment	Streamlined admission workflow
Accountant	Finance	Manage fees, receipts, refunds, concessions, and financial reports	Automated financial operations
Faculty	Academic Staff	Conduct classes, attendance, examinations, assignments, and grading	Simplified academic management
Reception Staff	Front Office	Handle visitor inquiries, student registrations, and general support	Faster information retrieval
Student	Learner	Attend classes, examinations, assignments, and monitor progress	Easy access to academic information
Parent / Guardian	Student Guardian	Monitor attendance, fees, academic performance, and institute communications	Real-time transparency and engagement
IT Administrator	Technical Support	System maintenance, backups, security, and infrastructure	Secure and reliable platform management

13. User Roles & Responsibilities

The Coaching Institute Management System implements Role-Based Access Control (RBAC) to ensure that every user accesses only the information and functionality necessary for their responsibilities.

Coaching Institute Management System (CIMS)- RAMP Project

The primary user roles include Super Administrator, Institute Owner, Managing Director, Branch Administrator, Accountant, Admission Counselor, Faculty Member, Reception Staff, Student, Parent, and System Support Engineer.

The Super Administrator possesses unrestricted system privileges including institute registration, subscription management, user provisioning, security configuration, system monitoring, audit management, backup administration, and platform maintenance.

The Institute Owner is responsible for overseeing organizational performance, monitoring financial reports, reviewing operational dashboards, approving strategic decisions, and supervising branch administration.

Branch Administrators manage branch-specific operations including student admissions, faculty allocation, timetable scheduling, classroom assignments, attendance verification, fee approvals, and branch reporting.

Faculty Members conduct academic activities including attendance recording through facial recognition, fingerprint authentication, or manual entry, examination management, assignment distribution, syllabus tracking, homework management, mark entry, and student performance evaluation.

Accountants manage fee collection, installment schedules, scholarship approvals, concessions, online payment reconciliation, financial reporting, refund processing, and outstanding fee recovery.

Admission Counselors register student inquiries, schedule counseling sessions, verify documents, complete admissions, assign batches, generate enrollment numbers, and coordinate onboarding activities.

Students can access their attendance records, examination schedules, assignments, marks, study materials, class timetables, announcements, fee information, and academic progress through secure portals.

Parents receive read-only access to attendance, fee status, examination results, homework, progress reports, timetable updates, institute announcements, and communication history relating exclusively to their enrolled children.

Every user action performed within the system shall be recorded within immutable audit logs to ensure accountability, traceability, security, and regulatory compliance

14. Functional Requirements

14.1 Authentication & Identity Management

The Authentication and Identity Management module shall provide secure access to all users based on Role-Based Access Control (RBAC).

Authentication Requirements

Req ID	Requirement
--------	-------------

Coaching Institute Management System (CIMS)- RAMP Project

FR-AUTH-001	The system shall allow users to authenticate using a unique username, registered email address, or registered mobile number.
FR-AUTH-002	The system shall support secure password authentication using encrypted password storage with BCrypt or Argon2 hashing algorithms.
FR-AUTH-003	The system shall support One-Time Password (OTP) authentication for password recovery and first-time account activation.
FR-AUTH-004	The system shall generate JSON Web Tokens (JWT) after successful authentication with configurable expiration times.
FR-AUTH-005	The system shall automatically invalidate expired authentication tokens and require re-authentication.
FR-AUTH-006	Multi-factor authentication (MFA) shall be supported for Super Administrators and Institute Owners.
FR-AUTH-007	User sessions shall automatically expire after a configurable period of inactivity.
FR-AUTH-008	The system shall prevent simultaneous login sessions beyond the configurable device limit.
FR-AUTH-009	Login attempts exceeding configurable thresholds shall temporarily lock the user account.
FR-AUTH-010	All authentication activities shall be recorded within immutable security audit logs.
FR-AUTH-011	Users shall be required to change system-generated passwords during their first successful login.
FR-AUTH-012	Passwords shall comply with configurable password complexity policies including uppercase letters, lowercase letters, numbers, and special characters.
FR-AUTH-013	The system shall provide secure password reset using OTP verification through SMS or Email.
FR-AUTH-014	User accounts shall support activation, deactivation, suspension, and permanent archival.
FR-AUTH-015	Authentication services shall support integration with Google OAuth and Microsoft authentication in future releases.

14.2 User & Role Management

The User Management module shall manage all users accessing the platform while Role Management shall define permissions based on organizational responsibilities.

User Management Requirements

Req ID	Requirement
FR-USER-001	The system shall allow administrators to create user accounts for all institute

Coaching Institute Management System (CIMS)- RAMP Project

	personnel.
FR-USER-002	Every user shall have a globally unique User ID generated automatically.
FR-USER-003	Each user account shall be associated with one or more organizational roles.
FR-USER-004	The system shall support role inheritance where higher-level roles automatically receive lower-level permissions where applicable.
FR-USER-005	Administrators shall assign branch-specific permissions to authorized users.
FR-USER-006	Every modification to user permissions shall be recorded in the audit log.
FR-USER-007	User profiles shall include personal information, contact details, designation, department, profile photograph, and digital signature where applicable.
FR-USER-008	The system shall support profile image upload with configurable size and format restrictions.
FR-USER-009	User accounts shall support activation, suspension, locking, unlocking, and archival without deleting historical records.
FR-USER-010	User search shall support filtering by name, role, department, branch, and account status.
FR-USER-011	The system shall maintain complete login history for every user.
FR-USER-012	The platform shall provide configurable password expiry policies.
FR-USER-013	User permissions shall be configurable without requiring application deployment.
FR-USER-014	The system shall support bulk user import using Excel and CSV templates.
FR-USER-015	Administrators shall receive notifications whenever privileged accounts are modified.

14.3 Institute Management

The Institute Management module shall manage organizational information, academic configuration, branding, branches, financial settings, and institutional preferences.

Institute Management Requirements

Req ID	Requirement
FR-INST-001	The system shall maintain complete institute profile information including institute name, registration number, address, contact details, website, logo, and branding information.
FR-INST-002	Multiple academic years shall be configurable without affecting historical records.
FR-INST-003	Academic calendars shall be configurable for each academic session.
FR-INST-004	The system shall support institute-wide configuration of working days, holidays, examination schedules, and operational timings.
FR-INST-005	SMS, Email, WhatsApp, Payment Gateway, Face Recognition, and Biometric configurations shall be manageable from centralized administrative settings.
FR-INST-006	Institute-wide notification templates shall be configurable.
FR-INST-007	Academic grading systems shall be configurable by administrators.
FR-INST-008	Digital certificate templates shall support institute branding and authorized signatures.

Coaching Institute Management System (CIMS)- RAMP Project

FR-INST-009	Backup schedules shall be configurable through administrative settings.
FR-INST-010	Institute administrators shall configure custom academic workflows without software modification.

14.4 Branch Management

The system shall support centralized management of multiple coaching branches operating under a single organization.

Branch Management Requirements

Req ID	Requirement
FR-BRANCH-001	The system shall support registration of multiple institute branches.
FR-BRANCH-002	Every branch shall maintain independent classrooms, faculty members, batches, students, and operational schedules.
FR-BRANCH-003	Branch administrators shall only access records belonging to their assigned branches.
FR-BRANCH-004	Institute Owners shall access consolidated reports across all branches.
FR-BRANCH-005	Branch performance dashboards shall display admissions, attendance, fee collections, examination statistics, and operational metrics.
FR-BRANCH-006	Student transfers between branches shall preserve complete academic history.
FR-BRANCH-007	Faculty transfers between branches shall maintain historical assignment records.
FR-BRANCH-008	Branch activation and deactivation shall not delete historical academic records.
FR-BRANCH-009	Branch-specific holidays shall override institute-wide holidays where configured.
FR-BRANCH-010	Branch reports shall support comparative analytics across all institute locations.

14.5 Student Management

The Student Management module shall manage the complete lifecycle of every student from inquiry to course completion.

Student Management Requirements

Req ID	Requirement
FR-STUDENT-001	The system shall generate a unique Student ID for every admitted student.
FR-STUDENT-002	Student profiles shall include personal information, parent information, previous academic history, emergency contacts, Aadhaar (optional), photograph, and uploaded documents.
FR-STUDENT-003	Students may enroll in multiple courses simultaneously.
FR-STUDENT-004	Student enrollment shall support online and offline admission workflows.
FR-STUDENT-005	Admission numbers shall be generated automatically based on configurable institute

Coaching Institute Management System (CIMS)- RAMP Project

	rules.
FR-STUDENT-006	Student profile photographs shall be captured for Face Recognition attendance registration.
FR-STUDENT-007	Fingerprint biometric registration shall support multiple fingerprints per student.
FR-STUDENT-008	Student document verification status shall be tracked throughout the admission process.
FR-STUDENT-009	Student transfers between batches shall preserve attendance, examination, and fee history.
FR-STUDENT-010	Student status shall include Active, Inactive, Completed, Suspended, Cancelled, and Alumni.
FR-STUDENT-011	Student search shall support advanced filtering using course, batch, branch, admission year, fee status, attendance percentage, and academic performance.
FR-STUDENT-012	The system shall maintain complete historical records without physical deletion of student information.
FR-STUDENT-013	Student profiles shall display attendance trends, fee summaries, examination performance, assignments, homework completion, and parent communication history.
FR-STUDENT-014	Student profile updates shall require authorization according to configured permission policies.
FR-STUDENT-015	Every modification to student records shall generate immutable audit entries.

14.6 Parent Portal

The Parent Portal shall provide real-time transparency regarding student academic activities.

Parent Portal Requirements

Req ID	Requirement
FR-PARENT-001	Parents shall securely authenticate using registered mobile numbers or email addresses.
FR-PARENT-002	Parents shall view real-time attendance immediately after attendance submission.
FR-PARENT-003	Parents shall receive notifications whenever students are absent.
FR-PARENT-004	Parents shall monitor fee payment history, outstanding balances, installment schedules payment receipts.
FR-PARENT-005	Parents shall access examination marks, rankings, report cards, assignments, homework progress reports.
FR-PARENT-006	Parents shall receive timetable modifications and institute announcements automatically.
FR-PARENT-007	Parents shall communicate with faculty members through secure messaging where permitted.
FR-PARENT-008	Parents shall download fee receipts, report cards, certificates, and attendance reports in PDF format.

Coaching Institute Management System (CIMS)- RAMP Project

	format.
FR-PARENT-009	Parents managing multiple children shall switch profiles without repeated authentication.
FR-PARENT-010	Parents shall receive emergency notifications immediately upon publication by the institute.

14.7 Course Management

The Course Management module shall enable administrators to define, organize, and maintain all academic programs offered by the institute. Each course shall support independent configurations including duration, syllabus, fee structure, faculty assignment, examination pattern, and certification requirements.

Course Management Requirements

Req ID	Requirement
FR-COURSE-001	The system shall allow administrators to create, modify, archive, and deactivate courses without affecting historical student records.
FR-COURSE-002	Every course shall have a unique Course Code generated automatically or configured manually.
FR-COURSE-003	Course information shall include course name, description, duration, eligibility criteria, fee structure, syllabus, maximum student capacity, and certification details.
FR-COURSE-004	A course may contain multiple subjects or modules with independent faculty assignments.
FR-COURSE-005	The system shall support online, offline, and hybrid course delivery modes.
FR-COURSE-006	Course fee structures shall support one-time payment, installment plans, scholarships, and concessions.
FR-COURSE-007	Multiple batches may be associated with the same course.
FR-COURSE-008	Course modifications shall not affect students already enrolled unless explicitly approved by authorized administrators.
FR-COURSE-009	Administrators shall define course prerequisites where applicable.
FR-COURSE-010	Course reports shall include enrollment statistics, completion rates, revenue, and student performance.

14.8 Batch Management

The Batch Management module shall organize students into academic groups based on courses, schedules, classrooms, faculty assignments, and branch locations.

Batch Management Requirements

Req ID	Requirement
FR-BATCH-001	The system shall allow creation of multiple batches for every course.

Coaching Institute Management System (CIMS)- RAMP Project

FR-BATCH-002	Each batch shall maintain a unique Batch ID and Batch Name.
FR-BATCH-003	Batch information shall include course, faculty, classroom, branch, start date, end date, capacity, and schedule.
FR-BATCH-004	Student enrollment shall automatically validate batch capacity before confirmation.
FR-BATCH-005	Students may be transferred between batches while preserving academic history.
FR-BATCH-006	Faculty assignments may be changed without affecting completed attendance and examination records.
FR-BATCH-007	Batch schedules shall support weekday, weekend, crash course, and customized recurring sessions.
FR-BATCH-008	Batch merging and splitting shall preserve all historical attendance and examination data.
FR-BATCH-009	Batch dashboards shall display attendance percentage, academic progress, fee collection, and examination analytics.
FR-BATCH-010	Batch completion shall automatically archive academic records while maintaining reporting accessibility.

14.9 Timetable Management

The Timetable Management module shall automate class scheduling while preventing faculty conflicts, classroom overlaps, and scheduling inconsistencies.

Timetable Requirements

Req ID	Requirement
FR-TIME-001	The system shall generate daily, weekly, and monthly timetables.
FR-TIME-002	Timetables shall automatically validate classroom availability before scheduling.
FR-TIME-003	Faculty scheduling conflicts shall be prevented automatically.
FR-TIME-004	The system shall detect overlapping classes and notify administrators before publishing schedules.
FR-TIME-005	Timetable modifications shall generate automatic notifications to students, parents, and faculty.
FR-TIME-006	Timetables shall support recurring schedules throughout the academic session.
FR-TIME-007	Special classes, revision sessions, seminars, and workshops shall be accommodated without affecting regular schedules.
FR-TIME-008	Holiday calendars shall automatically update timetable availability.
FR-TIME-009	Faculty leave shall automatically trigger timetable conflict alerts.
FR-TIME-010	Students shall view personalized timetables according to enrolled batches.

14.10 Attendance Management

The Attendance Management module shall provide multiple attendance recording mechanisms while maintaining high accuracy, security, and auditability.

Attendance shall support:

- Artificial Intelligence Face Recognition
- Biometric Fingerprint Authentication
- Manual Attendance Entry
- QR Code Attendance (Optional)
- RFID Card Attendance (Future Enhancement)

Attendance Requirements

Req ID	Requirement
FR-ATT-002	Attendance shall support Face Recognition verification using registered student facial profiles.
FR-ATT-003	Attendance shall support biometric fingerprint verification through integrated fingerprint devices.
FR-ATT-004	Authorized faculty shall manually record attendance when automated systems are unavailable.
FR-ATT-005	Attendance status shall include Present, Absent, Late, Half-Day, Excused, Medical Leave, and Holiday.
FR-ATT-006	Attendance timestamps shall be recorded automatically for every successful attendance transaction.
FR-ATT-007	Duplicate attendance entries for the same class session shall be prevented.
FR-ATT-008	Attendance correction requests shall require administrator approval.
FR-ATT-009	Attendance reports shall be generated daily, weekly, monthly, and annually.
FR-ATT-010	Parents shall receive configurable notifications when students are absent or arrive late.
FR-ATT-011	Attendance shall synchronize automatically across all authorized devices.
FR-ATT-012	Attendance records shall remain immutable after administrative approval except through authorized correction workflows.
FR-ATT-013	The system shall maintain complete attendance history throughout the student's academic lifecycle.
FR-ATT-014	Attendance analytics shall calculate attendance percentages automatically.
FR-ATT-015	Attendance dashboards shall identify chronically absent students using configurable thresholds.

14.11 Face Recognition Attendance

The Face Recognition subsystem shall utilize Artificial Intelligence to verify student identity before marking attendance.

Face Recognition Requirements

Req ID	Requirement
FR-FACE-001	Student facial profiles shall be registered during admission or profile creation.
FR-FACE-002	Multiple facial images shall be captured under different lighting conditions to improve recognition accuracy.
FR-FACE-003	The system shall perform real-time facial recognition before recording attendance.
FR-FACE-004	Liveness detection shall prevent spoofing using photographs or videos.
FR-FACE-005	Facial recognition confidence thresholds shall be configurable by administrators.
FR-FACE-006	Recognition failures shall prompt alternative attendance methods.
FR-FACE-007	All recognition attempts shall be logged for auditing purposes.
FR-FACE-008	Facial templates shall be securely encrypted within the database.
FR-FACE-009	Students may update facial profiles through authorized administrators.
FR-FACE-010	The system shall support future AI model upgrades without affecting stored attendance records.

14.12 Fingerprint Attendance

The Fingerprint Attendance module shall integrate with biometric devices for secure attendance verification.

Fingerprint Requirements

Req ID	Requirement
FR-BIO-001	The system shall support enrollment of multiple fingerprints per student.
FR-BIO-002	Fingerprint templates shall be encrypted before database storage.
FR-BIO-003	Attendance shall be recorded only after successful biometric verification.
FR-BIO-004	Fingerprint authentication failures shall trigger configurable retry attempts.
FR-BIO-005	Device connectivity failures shall automatically activate fallback attendance methods.
FR-BIO-006	Biometric devices from multiple vendors shall be supported through standardized APIs.
FR-BIO-007	Fingerprint synchronization shall occur automatically after network restoration.
FR-BIO-008	Device health monitoring shall notify administrators of hardware failures.

14.13 Examination Management

The Examination Management module shall automate examination planning, execution, evaluation, and academic reporting.

Examination Requirements

Req ID	Requirement
FR-EXAM-001	Administrators shall create examinations for one or multiple batches.
FR-EXAM-002	Examination schedules shall include date, time, duration, venue, and subject information.
FR-EXAM-003	Faculty members shall enter marks through secure digital interfaces.
FR-EXAM-004	Grading schemes shall be configurable for different courses.
FR-EXAM-005	The system shall calculate percentages, GPA, grades, and rankings automatically.
FR-EXAM-006	Absent students shall be handled separately within examination records.
FR-EXAM-007	Examination reports shall support PDF and Excel export.
FR-EXAM-008	Parents shall receive examination result notifications after publication.
FR-EXAM-009	Revaluation workflows shall maintain complete audit history.
FR-EXAM-010	Comparative performance analysis shall be generated across examinations.

14.14 Assignment & Homework Management

The Assignment module shall facilitate digital distribution, submission, evaluation, and feedback for academic assignments.

Assignment Requirements

Req ID	Requirement
FR-ASSIGN-001	Faculty shall create assignments for selected batches or courses.
FR-ASSIGN-002	Assignments shall support document, image, PDF, and video attachments.
FR-ASSIGN-003	Students shall submit assignments digitally before deadlines.
FR-ASSIGN-004	Late submissions shall be automatically identified.
FR-ASSIGN-005	Faculty shall evaluate assignments and publish marks with feedback.
FR-ASSIGN-006	Parents shall view assignment completion status.
FR-ASSIGN-007	Assignment history shall remain permanently accessible.
FR-ASSIGN-008	Assignment reminders shall be sent automatically before submission deadlines.

14.15 Study Material Management

The Study Material module shall provide centralized digital learning resources.

Coaching Institute Management System (CIMS)- RAMP Project

Study Material Requirements

Req ID	Requirement
FR-STUDY-001	Faculty shall upload notes, presentations, videos, PDFs, and practice materials.
FR-STUDY-002	Study materials shall be categorized by course, subject, chapter, and batch.
FR-STUDY-003	Students shall access only materials relevant to their enrolled courses.
FR-STUDY-004	Download permissions shall be configurable.
FR-STUDY-005	Version history shall be maintained for updated documents.
FR-STUDY-006	Access statistics shall identify the most frequently utilized learning resources.
FR-STUDY-007	Search functionality shall support keyword-based retrieval.
FR-STUDY-008	Faculty shall archive outdated learning materials without deleting historical versions.

14.16 Fee Management

The Fee Management module shall automate the complete financial lifecycle of student fee collection, including fee structures, installment plans, scholarships, concessions, online payments, refunds, and outstanding balance management.

Fee Management Requirements

Req ID	Requirement
FR-FEE-001	The system shall allow administrators to configure multiple fee structures for different courses, batches, and academic years.
FR-FEE-002	The system shall support one-time payments, monthly installments, quarterly installments, and customized payment plans.
FR-FEE-003	Fee structures shall support admission fees, tuition fees, examination fees, registration fees, study material fees, transport fees, hostel fees, and miscellaneous charges.
FR-FEE-004	Scholarships, discounts, concessions, and promotional offers shall be configurable independently.
FR-FEE-005	Fee invoices shall be generated automatically after student admission confirmation.
FR-FEE-006	The system shall calculate outstanding balances automatically after every payment transaction.
FR-FEE-007	Administrators shall define late fee penalties based on configurable business rules.
FR-FEE-008	Every payment shall generate a digitally signed receipt with a unique receipt number.
FR-FEE-009	The system shall maintain complete payment history throughout the student's academic lifecycle.
FR-FEE-010	Partial payments shall automatically update the remaining outstanding balance.
FR-FEE-011	Fee refunds shall require authorization according to institute approval policies.
FR-FEE-012	The system shall maintain complete audit trails for every financial transaction.

Coaching Institute Management System (CIMS)- RAMP Project

FR-FEE-013	Students shall not be physically deleted even after fee cancellation. Historical records shall remain accessible.
FR-FEE-014	Automatic fee reminders shall be generated before due dates and after overdue periods.
FR-FEE-015	Fee reports shall support filtering by course, batch, branch, academic year, payment status, and date range.

14.17 Online Payment Integration

The Online Payment module shall provide secure digital payment facilities for students and parents using multiple payment gateways.

Payment Gateway Requirements

Req ID	Requirement
FR-PAY-001	The system shall support UPI payments.
FR-PAY-002	The system shall support Debit Cards, Credit Cards, Net Banking, and Wallet payments.
FR-PAY-003	Payment gateway providers shall be configurable by administrators.
FR-PAY-004	Every successful transaction shall automatically generate a payment receipt.
FR-PAY-005	Failed transactions shall be automatically reconciled using payment gateway APIs.
FR-PAY-006	Parents shall receive payment confirmation through SMS, Email, and mobile notifications.
FR-PAY-007	Payment history shall remain permanently accessible.
FR-PAY-008	Refund transactions shall synchronize with the payment gateway automatically.
FR-PAY-009	Payment failures shall never create duplicate financial records.
FR-PAY-010	Sensitive payment information shall never be stored within the application database.

14.18 Notification & Communication Management

The Notification module shall provide centralized communication among administrators, faculty, students, and parents.

Notification Requirements

Req ID	Requirement
FR-NOTIFY-001	The system shall support SMS notifications.
FR-NOTIFY-002	The system shall support Email notifications.
FR-NOTIFY-003	The system shall support Push Notifications for mobile applications.
FR-NOTIFY-004	Future integration with WhatsApp Business API shall be supported.
FR-NOTIFY-005	Attendance notifications shall be sent immediately after attendance submission.

Coaching Institute Management System (CIMS)- RAMP Project

FR-NOTIFY-006	Fee reminder notifications shall be generated automatically before due dates.
FR-NOTIFY-007	Examination schedules and result announcements shall trigger automated notifications.
FR-NOTIFY-008	Timetable modifications shall notify affected students, faculty, and parents.
FR-NOTIFY-009	Emergency announcements shall be delivered with high priority.
FR-NOTIFY-010	Notification delivery history shall be maintained permanently.
FR-NOTIFY-011	Notification templates shall support dynamic placeholders.
FR-NOTIFY-012	Users shall configure individual notification preferences where permitted.

14.19 Dashboard & Business Analytics

The Dashboard module shall provide real-time analytical information tailored to each authorized user role.

Dashboard Requirements

Req ID	Requirement
FR-DASH-001	Institute Owners shall access organization-wide dashboards.
FR-DASH-002	Branch Managers shall access branch-specific dashboards.
FR-DASH-003	Faculty shall access class performance dashboards.
FR-DASH-004	Students shall access personal academic dashboards.
FR-DASH-005	Parents shall access child-specific dashboards.
FR-DASH-006	Dashboard widgets shall be configurable according to user roles.
FR-DASH-007	Admission trends shall be visualized using interactive charts.
FR-DASH-008	Revenue analytics shall display daily, monthly, quarterly, and yearly income.
FR-DASH-009	Attendance analytics shall display institute-wide and batch-wise attendance trends.
FR-DASH-010	Examination analytics shall display subject-wise and faculty-wise performance statistics.
FR-DASH-011	Outstanding fee analytics shall identify high-risk collections.
DASH-012	Dashboard data shall refresh automatically at configurable intervals.

14.20 Reports Management

The Reporting module shall generate operational, academic, administrative, and financial reports.

Report Requirements

Req ID	Requirement
FR-RPT-001	Student Master Report
FR-RPT-002	Admission Register
FR-RPT-003	Attendance Register

Coaching Institute Management System (CIMS)- RAMP Project

FR-RPT-004	Daily Attendance Summary
FR-RPT-005	Monthly Attendance Report
FR-RPT-006	Faculty Attendance Report
FR-RPT-007	Fee Collection Report
FR-RPT-008	Outstanding Fee Report
FR-RPT-009	Payment Receipt Register
FR-RPT-010	Examination Marks Register
FR-RPT-011	Student Progress Report
FR-RPT-012	Batch Performance Report
FR-RPT-013	Faculty Performance Report
FR-RPT-014	Branch Performance Report
FR-RPT-015	Revenue Analysis Report
FR-RPT-016	Scholarship & Concession Report
FR-RPT-017	Timetable Report
FR-RPT-018	Assignment Submission Report
FR-RPT-019	Parent Communication Report
FR-RPT-020	Audit Log Report

All reports shall support PDF, Excel, CSV, and print-friendly formats with configurable filters.

14.21 Certificate Management

The Certificate Management module shall automate certificate generation for students.

Certificate Requirements

Req ID	Requirement
FR-CERT-001	Course Completion Certificates shall be generated automatically after successful completion.
FR-CERT-002	Bonafide Certificates shall be generated by authorized administrators.
FR-CERT-003	Attendance Certificates shall calculate attendance percentages automatically.
FR-CERT-004	Certificates shall contain QR Codes for authenticity verification.
FR-CERT-005	Digital signatures shall be supported for institute authorities.
FR-CERT-006	Certificate templates shall support institute branding and customization.
FR-CERT-007	Certificate issuance history shall remain permanently available.

14.22 Artificial Intelligence Features (Future Ready)

The system architecture shall support Artificial Intelligence capabilities for future expansion.

AI Requirements

Req ID	Requirement
FR-AI-001	AI shall predict students at academic risk using attendance and examination history.
FR-AI-002	AI shall recommend personalized study plans based on student performance.
FR-AI-003	AI shall identify irregular attendance patterns automatically.
FR-AI-004	AI shall forecast fee collection trends.
FR-AI-005	AI shall provide institute performance analytics.
FR-AI-006	AI Chat Assistant shall answer frequently asked questions from students and parents.
FR-AI-007	AI shall assist administrators in generating analytical reports.
FR-AI-008	Future Generative AI integration shall support intelligent academic recommendations.

15. Business Rules

The following business rules govern system operations and ensure consistency across all modules.

Rule ID	Business Rule
BR-001	Every student shall have a unique Student ID.
BR-002	Every faculty member shall have a unique Employee ID.
BR-003	A student cannot be enrolled in an inactive course.
BR-004	Batch capacity shall not exceed configured limits without administrator approval.
BR-005	Attendance cannot be marked for cancelled classes.
BR-006	Duplicate attendance records for the same session shall not be permitted.
BR-007	Fee receipts shall have unique receipt numbers.
BR-008	Outstanding fee balances shall never become negative.
BR-009	Marks cannot exceed maximum examination marks.
BR-010	Parents shall only access information related to their linked children.
BR-011	Deleted users shall be archived instead of permanently removed.
BR-012	Every financial transaction shall be recorded within audit logs.
BR-013	Timetable conflicts shall prevent schedule publication until resolved.
BR-014	Face recognition attendance shall require successful liveness verification.
BR-015	Fingerprint authentication failures shall not automatically mark attendance.

16. Validation Rules

The system shall validate all user input before processing.

Validation ID	Validation Rule
VAL-001	Mobile numbers shall contain valid country-specific formats.
VAL-002	Email addresses shall comply with standard email validation rules.
VAL-003	Student age shall satisfy course eligibility requirements where configured.
VAL-004	Duplicate admission numbers shall not be permitted.
VAL-005	Fee payments shall not exceed outstanding balances unless authorized.
VAL-006	Attendance dates shall not exceed the current academic calendar.
VAL-007	Examination marks shall remain within configured grading limits.
VAL-008	Uploaded documents shall satisfy configured file size and format restrictions.
VAL-009	Biometric templates shall be verified before enrollment completion.
VAL-010	Face recognition enrollment shall require multiple facial samples meeting minimum lity thresholds.

17. Non-Functional Requirements

The Coaching Institute Management System shall satisfy the following non-functional requirements to ensure reliability, security, scalability, maintainability, and performance.

17.1 Performance Requirements

Req ID	Requirement
FR-PER-001	The system shall support a minimum of 5,000 active students per institute without performance degradation.
FR-PER-002	The application shall support at least 1,000 concurrent authenticated users.
FR-PER-003	Average API response time shall not exceed 2 seconds under normal operating conditions.
FR-PER-004	Dashboard pages shall load within 3 seconds for authenticated users.
FR-PER-005	Student search results shall be returned within 2 seconds.
FR-PER-006	Attendance submission shall complete within 2 seconds.
FR-PER-007	Face Recognition attendance verification shall complete within 3 seconds.
FR-PER-008	Fingerprint authentication shall complete within 2 seconds.
FR-PER-009	Report generation for datasets up to 50,000 records shall complete within 30 seconds.
FR-PER-010	System uptime shall be at least 99.9% excluding scheduled maintenance.

17.2 Scalability Requirements

Req ID	Requirement
FR-SCL-001	The system shall support multiple coaching institutes using a multi-tenant architecture if enabled.
FR-SCL-002	The platform shall support unlimited branches within a single institute.
FR-SCL-003	Additional application servers shall be added without application redesign.
FR-SCL-004	Database scaling shall support horizontal and vertical expansion.
FR-SCL-005	Cloud deployment shall support automatic resource scaling based on demand.

17.3 Availability Requirements

Req ID	Requirement
FR-AVL-001	Services shall remain available 24×7 except scheduled maintenance.
FR-AVL-002	Scheduled maintenance shall be announced at least 24 hours in advance.
FR-AVL-003	Automatic health monitoring shall detect service failures.
FR-AVL-004	Automatic restart mechanisms shall recover failed services whenever possible.
FR-AVL-005	Database failover mechanisms shall minimize downtime during infrastructure failures.

17.4 Reliability Requirements

Req ID	Requirement
NFR-REL-001	Student records shall never be permanently deleted through standard operations.
NFR-REL-002	All financial transactions shall follow ACID database principles.
NFR-REL-003	Attendance records shall be protected against accidental modification.
NFR-REL-004	Automatic transaction rollback shall occur during processing failures.
NFR-REL-005	Every critical operation shall generate audit records.

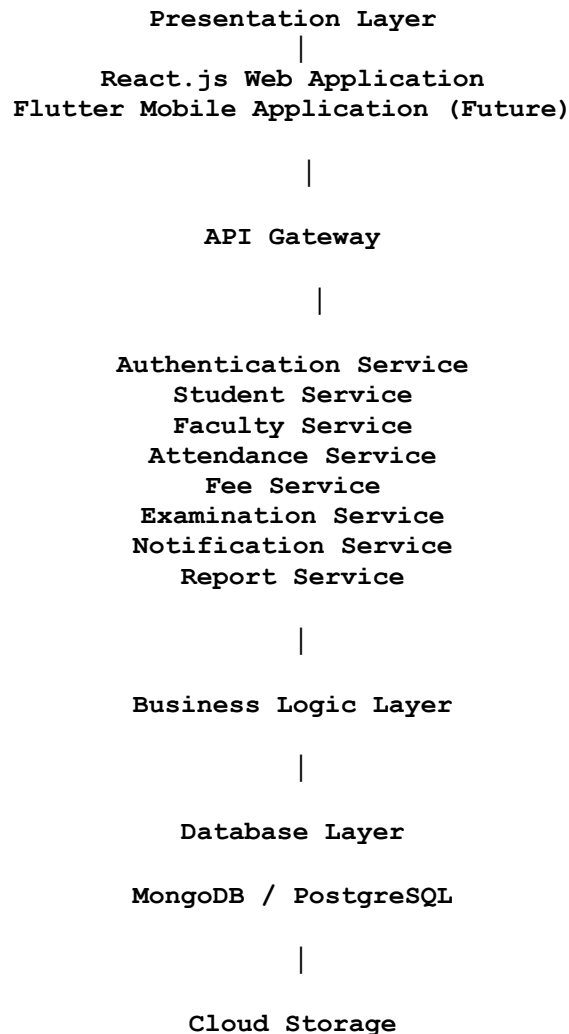
17.5 Usability Requirements

Req ID	Requirement
NFR-USE-001	The user interface shall remain responsive across desktop, tablet, and mobile devices.
NFR-USE-002	Navigation shall be role-specific and intuitive.
NFR-USE-003	Error messages shall provide meaningful guidance to users.
NFR-USE-004	Frequently used operations shall require minimal user interaction.
NFR-USE-005	Accessibility guidelines shall be considered during interface design.

18. System Architecture

The Coaching Institute Management System shall adopt a modern layered architecture based on modular services to ensure maintainability, scalability, security, and future extensibility.

Architectural Layers



Technology Stack

Layer	Technology
Frontend	React.js, TypeScript, Tailwind CSS
Mobile	Flutter (Future Release)
Backend	Node.js, Express.js
Authentication	JWT + Refresh Tokens
Database	MongoDB or PostgreSQL

Coaching Institute Management System (CIMS)- RAMP Project

Cache	Redis
Storage	AWS S3 / Cloudinary
API	REST APIs
Face Recognition	FaceNet / TensorFlow / OpenCV
Fingerprint SDK	Vendor Device SDK
Reports	JasperReports / PDF Generator
Deployment	Docker + Nginx
Cloud	AWS / Azure / GCP

19. Database Design

The system shall maintain a normalized relational or document database depending on deployment architecture.

Primary Database Entities

Entity ID	Entity
DB-001	Institute
DB-002	Branch
DB-003	User
DB-004	Role
DB-005	Permission
DB-006	Student
DB-007	Parent
DB-008	Faculty
DB-009	Employee
DB-010	Course
DB-011	Subject
DB-012	Batch
DB-013	Classroom
DB-014	Timetable
DB-015	Attendance
DB-016	Face Recognition Profile
DB-017	Fingerprint Profile
DB-018	Fee Structure
DB-019	Fee Invoice
DB-020	Payment

Coaching Institute Management System (CIMS)- RAMP Project

DB-021	Receipt
DB-022	Scholarship
DB-023	Examination
DB-024	Marks
DB-025	Grade
DB-026	Assignment
DB-027	Homework
DB-028	Study Material
DB-029	Notification
DB-030	SMS Queue
DB-031	Email Queue
DB-032	Audit Log
DB-033	System Configuration
DB-034	Holiday Calendar
DB-035	Academic Year
DB-036	Announcement
DB-037	Certificate
DB-038	Activity Log
DB-039	Login History
DB-040	Backup History

20. API Requirements

The application shall expose secure RESTful APIs for all client applications.

Authentication APIs

API ID	Endpoint
API-AUTH-001	POST /api/auth/login
API-AUTH-002	POST /api/auth/logout
API-AUTH-003	POST /api/auth/register
API-AUTH-004	POST /api/auth/refresh-token
API-AUTH-005	POST /api/auth/forgot-password

Student APIs

API ID	Endpoint
API-STU-001	GET /api/students

Coaching Institute Management System (CIMS)- RAMP Project

API-STU-002	GET /api/students/{id}
API-STU-003	POST /api/students
API-STU-004	PUT /api/students/{id}
API-STU-005	DELETE /api/students/{id}

Attendance APIs

API ID	Endpoint
API-ATT-001	POST /api/attendance
API-ATT-002	GET /api/attendance
API-ATT-003	POST /api/attendance/face
API-ATT-004	POST /api/attendance/fingerprint
API-ATT-005	POST /api/attendance/manual

Fee APIs

API ID	Endpoint
API-FEE-001	POST /api/fees
API-FEE-002	POST /api/payments
API-FEE-003	GET /api/invoices
API-FEE-004	GET /api/receipts

Report APIs

API ID	Endpoint
API-RPT-001	GET /api/reports/attendance
API-RPT-002	GET /api/reports/fees
API-RPT-003	GET /api/reports/examinations
API-RPT-004	GET /api/reports/dashboard

21. Security Requirements

Req ID	Requirement
SEC-001	All communication shall occur exclusively over HTTPS.
SEC-002	JWT access tokens shall be digitally signed and validated before every request.
SEC-003	Passwords shall be hashed using BCrypt or Argon2.

Coaching Institute Management System (CIMS)- RAMP Project

- SEC-004 Sensitive personal information shall be encrypted at rest using AES-256.
- SEC-005 Role-Based Access Control (RBAC) shall be enforced for every API endpoint.
- SEC-006 SQL Injection, NoSQL Injection, and XSS attacks shall be prevented through input validation and parameterized queries.
- SEC-007 Cross-Site Request Forgery (CSRF) protection shall be enabled where applicable.
- SEC-008 Security headers including HSTS, CSP, X-Frame-Options, and X-Content-Type-Options shall be configured.
- SEC-009 Login attempts shall be rate limited to prevent brute-force attacks.
- SEC-010 All privileged activities shall be recorded within immutable audit logs.
- SEC-011 Uploaded documents shall undergo antivirus and malware scanning before storage.
- SEC-012 Face recognition templates and biometric fingerprint templates shall be encrypted and inaccessible to unauthorized users.

22. Audit Requirements

Audit ID Requirement

- AUD-001 Every login attempt shall be recorded.
- AUD-002 Student profile modifications shall generate audit records.
- AUD-003 Attendance corrections shall record previous and updated values.
- AUD-004 Fee modifications shall require audit approval.
- AUD-005 Examination mark modifications shall preserve historical values.
- AUD-006 Administrative configuration changes shall be permanently logged.
- AUD-007 Every report generation request shall be recorded.
- AUD-008 Audit records shall be retained according to configurable retention policies.

23. Integration Requirements

Integration	Description
Face Recognition Engine	AI-powered facial attendance verification
Fingerprint Scanner SDK	Biometric attendance integration
Payment Gateway	Razorpay, PhonePe, Stripe, PayPal (configurable)
SMS Gateway	OTP and institute notifications
Email Service	Password recovery and communications
Push Notification Service	Mobile application notifications
WhatsApp Business API	Future communication integration
Google Calendar	Timetable synchronization (future)
Zoom / Google Meet	Online classes (future)
Cloud Storage	Secure storage for study materials and documents

24. Backup & Disaster Recovery

Requirement ID	Requirement
BDR-001	Full database backup shall be performed daily.
BDR-002	Incremental backups shall be performed every hour.
BDR-003	Backup files shall be encrypted before cloud storage.
BDR-004	Disaster recovery procedures shall support restoration within four hours.
BDR-005	Backup integrity shall be verified automatically after every backup cycle.

25. Deployment Architecture

The application shall support cloud-native deployment using containerized services.

Internet Users

↓

Cloud Load Balancer

↓

Coaching Institute Management System (CIMS)- [RAMP Project](#)

Nginx Reverse Proxy

↓

React Frontend

↓

Node.js Backend APIs

↓

Redis Cache

↓

MongoDB / PostgreSQL

↓

AWS S3 Storage

↓

Backup Server

Deployment shall support development, testing, staging, and production environments with independent configurations and automated CI/CD pipelines.

26. Data Migration Strategy

The Coaching Institute Management System shall support a structured, secure, and verifiable migration process to transfer data from existing manual records, spreadsheets, or legacy software into the new platform. The migration process shall ensure data integrity, minimize operational disruption, and preserve historical records.

26.1 Migration Objectives

- Transfer all existing student records into the new system.
- Import faculty, staff, and parent information.
- Migrate course, batch, and timetable details.
- Import fee history, payment records, and outstanding balances.
- Preserve attendance records wherever available.
- Maintain examination history and academic performance.
- Validate imported data before production deployment.

26.2 Data Migration Phases

Phase	Description
Planning	Identify data sources, define migration scope, prepare migration strategy.
Data Extraction	Extract records from Excel, CSV, databases, or legacy applications.
Data Cleansing	Remove duplicate, incomplete, and inconsistent records.
Data Transformation	Convert data into the CIMS database structure.
Validation	Verify imported data for completeness and accuracy.
Pilot Migration	Perform migration using sample data.
Final Migration	Import production data after approval.
Verification	Validate migrated records with stakeholders.
Go-Live	Enable production system after successful verification.

26.3 Supported Import Formats

- Microsoft Excel (.xlsx)
- CSV
- SQL Database Backup
- JSON
- XML (Optional)

27. Acceptance Criteria

The Coaching Institute Management System shall be considered successfully implemented when the following acceptance criteria are satisfied.

Functional Acceptance

- All approved functional requirements have been implemented.
- All user roles operate according to defined permissions.
- Student admission workflow operates successfully.
- Attendance recording functions correctly using supported methods.
- Fee collection and receipt generation work without errors.
- Examination and report card generation operate correctly.
- Notifications are delivered successfully.
- Dashboards display accurate information.
- Reports generate successfully.

Performance Acceptance

- API response time meets specified limits.
- Concurrent user support satisfies performance requirements.
- Database transactions execute without inconsistencies.
- Face Recognition attendance completes within acceptable time limits.
- System remains available during peak operational hours.

Security Acceptance

- Role-Based Access Control functions correctly.
- Unauthorized access attempts are rejected.
- Sensitive information is encrypted.
- Audit logs capture all critical activities.
- Authentication and authorization operate securely.

28. User Acceptance Testing (UAT)

The following representative User Acceptance Test scenarios shall be executed before production deployment.

Test ID	Case Scenario	Expected Result
UAT-001	Student Admission	Student record created successfully.
UAT-002	Batch Assignment	Student assigned to selected batch.

Coaching Institute Management System (CIMS)- RAMP Project

UAT-003	Manual Attendance	Attendance saved correctly.
UAT-004	Face Recognition Attendance	Student identified and attendance recorded.
UAT-005	Fingerprint Attendance	Attendance recorded after biometric verification.
UAT-006	Fee Payment	Payment processed and receipt generated.
UAT-007	Online Payment	Payment gateway transaction completed successfully.
UAT-008	Examination Marks Entry	Marks saved and calculated correctly.
UAT-009	Report Card Generation	Report card generated accurately.
UAT-010	Parent Login	Parent accesses only linked student information.
UAT-011	Student Dashboard	Student views attendance, marks, fees, and timetable.
UAT-012	Faculty Dashboard	Faculty manages attendance and examinations.
UAT-013	Timetable Update	Notifications sent after timetable modification.
UAT-014	Attendance Notification	Parent receives absence notification.
UAT-015	Report Generation	Reports exported successfully in PDF and Excel formats.

29. Risk Analysis

The project team shall continuously identify, evaluate, monitor, and mitigate risks throughout the software development lifecycle.

Risk ID	Risk	Impact	Mitigation
RISK-001	Data migration errors	High	Perform pilot migration and verification.
RISK-002	Server downtime	High	High-availability deployment and monitoring.
RISK-003	Payment gateway failure	High	Automatic retries and reconciliation.
RISK-004	Face recognition inaccuracies	Medium	Quality enrollment, configurable confidence threshold, fallback attendance.
RISK-005	Fingerprint device malfunction	Medium	Manual attendance fallback.
RISK-006	Unauthorized access	High	RBAC, MFA, encryption, security monitoring.
RISK-007	Data loss	High	Automated backups and disaster recovery.
RISK-	Low user adoption	Medium	User training and support documentation.

008

RISK- Network failures Medium Offline data capture where applicable and automatic synchronization.

RISK- Third-party API Medium Retry mechanisms and graceful degradation.
010 failures

30. Assumptions and Constraints

Assumptions

- Users possess valid login credentials.
- Internet connectivity is available for cloud-hosted deployments.
- Biometric devices and cameras are installed where biometric attendance is required.
- Institute administrators configure courses, batches, fee structures, and academic calendars before operations begin.
- Payment gateway accounts and messaging services are configured by the institute.
- Required hardware meets the minimum system requirements.

Constraints

- Face Recognition accuracy depends on camera quality, lighting, and enrolled facial data.
- Fingerprint authentication depends on compatible biometric devices.
- Third-party service availability (SMS, Email, Payment Gateway, WhatsApp API) is outside application control.
- Performance depends on network bandwidth and server capacity.
- Government regulations may require future compliance updates.

31. Training Requirements

The successful implementation of the system requires structured training for all stakeholder groups.

User Group	Training Topics
Super Administrator	System configuration, security, user management, backup, monitoring
Institute Owner Administrator	Dashboards, analytics, reports, branch monitoring
Branch Manager	Admissions, fee management, attendance, examinations, reporting
Faculty	Branch operations, student management, scheduling
Accountant	Attendance, assignments, examinations, marks, study materials
Students	Fee collection, invoices, receipts, refunds, reports
	Dashboard, timetable, assignments, study materials

Parents Attendance, fees, progress reports, notifications

Training materials shall include:

- User Manuals
- Administrator Guide
- Video Tutorials
- Quick Reference Guides
- Frequently Asked Questions (FAQ)
- Standard Operating Procedures (SOP)

32. Warranty and Maintenance

Following successful deployment, the software shall be maintained according to the agreed maintenance policy.

Warranty Period

- Bug fixes for functional defects.
- Security patches.
- Performance optimization.
- Database maintenance.
- Minor UI improvements.

Preventive Maintenance

- Database optimization.
- Backup verification.
- Security updates.
- Log cleanup.
- Performance monitoring.
- Infrastructure health checks.

Corrective Maintenance

- Bug resolution.
- Error correction.
- Incident management.
- Data recovery support.

Adaptive Maintenance

- Operating system compatibility.
- Browser compatibility updates.

Coaching Institute Management System (CIMS)- RAMP Project

- Third-party API updates.
- Regulatory compliance changes.

Perfective Maintenance

- Feature enhancements.
- User experience improvements.
- Dashboard enhancements.
- Performance optimization.

33. Future Enhancements

The architecture has been designed to support future expansion without major redevelopment.

Potential future enhancements include:

- AI-based student performance prediction.
- AI-generated personalized study recommendations.
- AI-powered chatbot for students and parents.
- Online live classes.
- Learning Management System (LMS) integration.
- Digital library.
- Online examination with AI-based proctoring.
- Hostel management.
- Transport management with GPS tracking.
- Library management.
- Inventory management.
- Payroll management.
- Human Resource Management.
- Visitor Management System.
- QR Code student identification.
- RFID Smart ID Cards.
- WhatsApp Business integration.
- Mobile applications for Android and iOS.
- Voice-enabled attendance.
- Face Recognition attendance using mobile devices.
- Predictive admission analytics.
- Data warehouse and business intelligence dashboards.
- Multi-language interface.
- Multi-tenant SaaS deployment.

34. Requirements Traceability Matrix (RTM)

Business Requirement	Functional Requirement	Module	Test Case
Student Admission	FR-STUDENT-001 to FR-STUDENT-015	Student Management	UAT-001
Authentication	FR-AUTH-001 to FR-AUTH-015	Authentication	UAT-010
Attendance	FR-ATT-001 to FR-ATT-015	Attendance	UAT-003
Face Recognition	FR-FACE-001 to FR-FACE-010	Attendance	UAT-004
Fingerprint	FR-BIO-001 to FR-BIO-008	Attendance	UAT-005
Fee Collection	FR-FEE-001 to FR-FEE-015	Finance	UAT-006
Online Payments	FR-PAY-001 to FR-PAY-010	Finance	UAT-007
Examination	FR-EXAM-001 to FR-EXAM-010	Examination	UAT-008
Parent Portal	FR-PARENT-001 to FR-PARENT-010	Parent Portal	UAT-010
Dashboard	FR-DASH-001 to FR-DASH-012	Analytics	UAT-011
Reporting	FR-RPT-001 to FR-RPT-020	Reports	UAT-015

35. Appendix

A. Recommended Hardware Requirements

Server

- 8-Core CPU or higher
- 32 GB RAM
- 500 GB SSD
- Linux Server (Ubuntu LTS recommended)

Client Systems

- Intel Core i3 or higher
- 8 GB RAM
- Modern Web Browser (Chrome, Edge, Firefox, Safari)
- Stable Internet Connection

Biometric Devices

- Face Recognition Camera (HD/IP Camera or supported webcam)
- Fingerprint Scanner (SDK-supported device)
- QR Code Scanner (Optional)

B. Recommended Software Stack

Component	Technology
Frontend	React.js, TypeScript, Tailwind CSS
Mobile	Flutter (Future)
Backend	Node.js, Express.js
Database	PostgreSQL or MongoDB
Cache	Redis
Authentication	JWT + Refresh Tokens
Storage	AWS S3 / Cloudinary
Reporting	PDF Generator, Excel Export
DevOps	Docker, Nginx, GitHub Actions
Monitoring	Prometheus, Grafana
Logging	ELK Stack / Loki

C. Glossary

Term Meaning

CIMS	Coaching Institute Management System
RBAC	Role-Based Access Control
JWT	JSON Web Token
OTP	One-Time Password
API	Application Programming Interface
AI	Artificial Intelligence
KPI	Key Performance Indicator
UAT	User Acceptance Testing
RTM	Requirements Traceability Matrix
MFA	Multi-Factor Authentication
SaaS	Software as a Service

Document Approval

Role	Responsibility	Status
Business Analyst	Requirements Definition	Pending
Solution Architect	Technical Review	Pending
Development Lead	Implementation Review	Pending

Coaching Institute Management System (CIMS)- RAMP Project

QA Lead	Test Strategy Review	Pending
Project Manager	Project Approval	Pending
Institute Owner / Client	Final Acceptance	Pending

Document Control

- **Document Title:** Software Requirements Specification (SRS)
- **Project Name:** Coaching Institute Management System (CIMS)
- **Version:** 1.0
- **Document Status:** Draft
- **Prepared By:** Software Engineering Team
- **Reviewed By:** Project Manager & Technical Architect
- **Approved By:** Client / Institute Management
- **Confidentiality:** Internal Use Only